Timing Search[1]

Online Appendix

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A One-Sided Timing Search

The one-sided timing search problem considers an agent who observes market conditions at Poisson rate λ and chooses when to act. With i.i.d. payoffs, the reservation value satisfies the standard McCall equation

$$rv^* = -c + \lambda \int_{v^*}^{\bar{v}} (v - v^*)\, dF(v),$$

and timing search is isomorphic to cross-sectional search.

Under Ornstein–Uhlenbeck payoffs ($dv = \theta(\bar{v} - v)dt + \sigma\, dW$), two results emerge with no cross-sectional analog. First, the reservation value depends on mean-reversion speed θ: $\partial v^*/\partial\theta < 0$ when $v^* > \bar{v}$ and > 0 when $v^* < \bar{v}$. Second, the option value satisfies $\frac{\partial}{\partial v}[W(v) - v] < -1$ for $v \gg \bar{v}$ (agents rush to act at high payoffs), versus exactly -1 under i.i.d. search.

B Matching Microfoundation for Gaussian Profits

This appendix provides a matching-theoretic microfoundation for the Gaussian matching efficiency function $m(n) = \exp(-\frac{\varphi}{2}(n - n_p)^2)$ stated in Assumption 2 of the main text (Gaussian scale efficiency).

Consider a market with n active firms. Each potential match between a firm and a trading partner has a quality θ drawn from a normal distribution $\theta \sim N(n_p, 1/\varphi)$. A firm operating at market thickness n captures matches with quality near n: specifically, matches within distance ε of n on the quality dimension. The density of good matches available to the firm is proportional to the normal density evaluated at n:

$$m(n) \propto \frac{1}{\sqrt{2\pi/\varphi}} \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big).$$

This is the Hotelling–Salop logic applied to market thickness: firms compete for matches along the density dimension, and the normal distribution generates the Gaussian kernel. Absorbing the normalizing constant into π_0 yields $\pi(n) = \pi_0 \exp(-\frac{\varphi}{2}(n - n_p)^2)$.

The parameter φ inherits its interpretation from the variance of match qualities: $\varphi = 1/\mathrm{Var}(\theta)$. Industries with homogeneous match qualities (low variance, high φ)

have a narrow peak and are sensitive to scale deviations. Industries with heterogeneous qualities (high variance, low φ) have a broad peak and are robust to deviations.

C Derivation of the Timing Friction

This appendix derives the exponential-quadratic timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ as the continuous-time limit of proportional adjustment costs.

Discrete-time formulation. Consider a discrete-time economy with period length $\Delta > 0$. In each period, effective output is reduced by a fraction proportional to the square of the adjustment speed:

$$y_t = \pi(n_t) \cdot \left[1 - \mu \left(\frac{n_{t+\Delta} - n_t}{\Delta}\right)^2 \Delta\right].$$

The adjustment cost is proportional to the square of the rate of change $(\Delta n/\Delta)^2$, scaled by the period length Δ. The parameter $\mu > 0$ governs the severity of the friction.

Continuous-time limit. As $\Delta \to 0$, write $\dot{n} = \lim_{\Delta\to 0}(n_{t+\Delta} - n_t)/\Delta$ and note that $1 - x \approx e^{-x}$ for small x. The effective output becomes:

$$y(t) = \pi(n(t)) \cdot e^{-\frac{\mu}{2}\dot{n}(t)^2} = \pi(n(t)) \cdot \xi(\dot{n}(t)).$$

Robustness of the quadratic form. Any symmetric friction $\xi(\dot{n})$ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$, where $\mu = -\xi''(0)$. The leading-order dynamics—in particular $\omega = \sqrt{\varphi/\mu}$—depend only on μ, not on higher-order terms. The exponential-quadratic form is therefore canonical.

D Proof of Proposition 6 (Stochastic Limit Cycle)

This appendix provides the proof of Proposition 6 in the main text, which states that the spectral density of $n(t)$ under small stochastic forcing peaks at $\omega = \sqrt{\varphi/\mu}$.

Action–angle coordinates. Write the deterministic solution as $n(t) = n_p + A\cos\phi(t)$ and $\dot{n}(t) = -A\omega\sin\phi(t)$, where $\phi(t) = \omega t + \phi_0$ and $\omega = \sqrt{\varphi/\mu}$. Transform the stochastic system to action–angle coordinates (A, ϕ).

Under the perturbation $\tilde{\pi} = \pi + \sigma_\varepsilon \varepsilon_t$, the energy $E = -\frac{\varphi}{2}A^2$ evolves as:

$$\mathrm{d}E = \frac{\sigma_\varepsilon \varepsilon_t}{\pi(n)} \dot{n}\,\mathrm{d}t.$$

Since $\dot{n} = -A\omega\sin\phi$ and $\pi(n) \approx \pi_0 e^E$ near the equilibrium orbit:

$$\mathrm{d}A \approx -\frac{\sigma_\varepsilon}{\varphi A \pi_0 e^E} A\omega\sin\phi\,\varepsilon_t\,\mathrm{d}t.$$

Averaging over the fast phase ϕ, the amplitude A evolves as a diffusion with drift coefficient determined by the noise coupling.

Spectral density. The process $n(t)$ is a stochastically forced harmonic oscillator:

$$\ddot{n} + \omega^2(n - n_p) = \text{noise},$$

where the noise term inherits the i.i.d. structure of ε_t. The transfer function of this oscillator is:

$$H(\nu) = \frac{1}{\omega^2 - \nu^2 + i\eta\nu},$$

where η is an effective damping coefficient arising from the amplitude diffusion. The power spectral density is $S(\nu) = |H(\nu)|^2 S_\varepsilon(\nu)$, which peaks at $\nu = \omega = \sqrt{\varphi/\mu}$. Since S_ε is flat (white noise), the peak location is determined entirely by ω, not by σ_ε.

Properties.

(i) The spectral peak is at $\omega = \sqrt{\varphi/\mu}$, as claimed.

(ii) The peak location depends on (φ, μ) only; the shock variance σ_ε^2 affects the peak *height* and *width*, not its location.

(iii) The persistence of the cycle is endogenous: even i.i.d. shocks produce a spectral peak because the transfer function $|H(\nu)|^2$ concentrates power near ω. ■

Role of shock structure. Shock variance σ_ε^2 scales the peak height and width without shifting its location, since $|H(\nu)|^2$ peaks at $\nu = \omega$ regardless of the level of S_ε. With persistent shocks, $S_\varepsilon(\nu)$ is no longer flat, and the observed spectral density $S(\nu) = |H(\nu)|^2 S_\varepsilon(\nu)$ peaks where the product is maximized. Since $|H(\nu)|^2$ peaks at ω while a persistent process (e.g., AR(1)) tilts S_ε toward low frequencies, the observed peak can shift slightly below ω. The displacement is of order ρ/Q^2, where ρ is the shock autocorrelation and $Q = \omega/\eta$ is the quality factor of the transfer function. For sharp peaks ($Q \gg 1$), the displacement is negligible.

E Proof of Proposition 3 (General Hump-Shaped Profits)

Proof. Let $\pi(n)$ be a C^2 function with $\pi(n) > 0$ for n in a neighborhood of n_p, $\pi'(n_p) = 0$, and $\pi''(n_p) < 0$. The ironing condition gives:

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E.$$

Define the potential $U(n) = -\frac{1}{\mu}\ln \pi(n)$, which captures how the profit landscape confines the economy near optimal scale. Since $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, we have $U'(n_p) = 0$ and $U''(n_p) = -\pi''(n_p)/(\mu\pi(n_p)) > 0$, so n_p is a local minimum of U. This becomes:

$$\frac{1}{2}\dot{n}^2 + U(n) = -\frac{E}{\mu} \equiv \mathcal{E}.$$

Economically, this says that the sum of the profit-landscape cost $U(n)$ and the adjustment cost $\frac{1}{2}\dot{n}^2$ is constant along the equilibrium path. The economy oscillates around n_p because deviations from optimal scale are offset by adjustment dynamics. For $\mathcal{E}$ slightly above $U(n_p)$, the oscillations have small amplitude. The frequency of small oscillations is:

$$\omega = \sqrt{U''(n_p)} = \sqrt{\frac{-\pi''(n_p)}{\mu\,\pi(n_p)}} = \sqrt{\frac{|f'(n_p)|}{\mu}},$$

where $f'(n_p) = [\pi''/\pi - (\pi'/\pi)^2]|_{n_p} = \pi''(n_p)/\pi(n_p)$ since $\pi'(n_p) = 0$.

For finite amplitude A, the period of oscillation under a general profit landscape

is:

$$T(A) = \sqrt{2} \int_{n_-}^{n_+} \frac{\mathrm{d}n}{\sqrt{\mathcal{E} - U(n)}},$$

where $n_\pm$ are the turning points. Expanding $U(n)$ to fourth order around n_p:

$$U(n) = U(n_p) + \frac{1}{2}U''(n_p)(n - n_p)^2 + \frac{1}{6}U'''(n_p)(n - n_p)^3 + \frac{1}{24}U^{(4)}(n_p)(n - n_p)^4 + \cdots$$

Standard perturbation theory for anharmonic oscillators gives:

$$\omega(A) = \omega_0 + O(A^2),$$

where $\omega_0 = \sqrt{U''(n_p)} = \sqrt{|f'(n_p)|/\mu}$.

For the Gaussian specification, $U(n) = \frac{\varphi}{2\mu}(n - n_p)^2 + \text{const}$ exactly (all higher-order terms vanish), so $\omega = \omega_0$ exactly for all A. For $\pi(n) = n^\alpha(a - bn)$, the cubic and quartic corrections are nonzero, producing amplitude-dependent corrections to the frequency. ■

F Entry at All Dates: Verification of the Regularity Condition

This appendix verifies that the regularity condition "entry occurs at all dates" in Proposition 2 is satisfied under the stated conditions.

Proposition 1.
Suppose $M > n_p + A$ (the total population exceeds the maximum of $n(t)$ along the cycle). Then in the timing search equilibrium of Theorem 1, the searcher pool satisfies $S(t) = M - n(t) \geq M - n_p - A > 0$ for all t, and entry occurs at all dates.

Proof. Along the equilibrium path, $n(t) = n_p + A\cos(\omega t + \phi) \leq n_p + A < M$. Therefore $S(t) = M - n(t) \geq M - n_p - A > 0$ for all t.

It remains to show that $e(t) > 0$ for all t. From the aggregate consistency condition, $e(t) = \dot{n}(t) + \delta n(t)$. Using $n(t) = n_p + A\cos(\omega t + \phi)$:

$$e(t) = -A\omega\sin(\omega t + \phi) + \delta(n_p + A\cos(\omega t + \phi)).$$

For $e(t) > 0$ to hold at all dates, we need $\delta n_p > A\sqrt{\omega^2 + \delta^2}$, i.e., the exit flow

δn_p must exceed the maximum amplitude of the net flow $\dot{n} + \delta A \cos(\cdot)$. This is satisfied whenever the cycle amplitude is moderate relative to the steady-state scale: $A < \delta n_p / \sqrt{\omega^2 + \delta^2}$. ■

G Detailed Derivation of Entry-Matching Micro-Foundation

This appendix provides the detailed derivation of Proposition 1 in the main text (micro-foundation of the timing friction from entry matching congestion).

Setup. Under Assumption 3, the matching function is $m(S, R) = S^\alpha R^{1-\alpha}$, where $S = M - n$ is the searcher pool and R is the stock of entry resources. The entry flow is $e(t) = S(t) \cdot p(\theta_e(t))$, where $p(\theta_e) = \theta_e^{-(1-\alpha)}$ and $\theta_e = S/R$.

Steady state. At steady state, $\dot{n} = 0$, so $e_{ss} = \delta n_p$. The steady-state tightness is $\theta_e^{ss} = (M - n_p)/R$.

Perturbation. When $\dot{n} \neq 0$, the entry flow must accommodate both replacement of exiting firms and net entry: $e = \delta n + \dot{n}$. Around steady state ($n \approx n_p$, $\dot{n}$ small):

$$e = \delta n_p + \dot{n} + \delta(n - n_p) \approx \delta n_p + \dot{n},$$

to leading order in perturbations. The matching probability depends on the flow through the tightness it implies. Specifically, the entry rate per resource unit is $q(\theta_e) = \theta_e^\alpha$, so $e = R \cdot q(\theta_e)$, giving $\theta_e = (e/R)^{1/\alpha}$.

The matching probability per searcher is:

$$p(\theta_e) = \theta_e^{-(1-\alpha)} = \left(\frac{e}{R}\right)^{-(1-\alpha)/\alpha}.$$

Effective friction. The effective output friction from entry congestion is:

$$\ln \xi(\dot{n}) \equiv \ln p(\theta_e(\dot{n})) - \ln p(\theta_e^{ss}) = -\frac{1-\alpha}{\alpha}\left[\ln\left(\frac{e}{e_{ss}}\right)\right] = -\frac{1-\alpha}{\alpha}\ln\left(1 + \frac{\dot{n}}{\delta n_p}\right).$$

Taylor expansion. Let $z = \dot{n}/(\delta n_p)$. Then:

$$\ln \xi = -\frac{1-\alpha}{\alpha}\left[z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots\right].$$

The linear term $-(1-\alpha)z/\alpha$ is a first-order correction that shifts the equilibrium level. The constant C in the ironing condition adjusts to absorb this shift. Define $\tilde{\xi}(\dot{n}) = \xi(\dot{n}) \cdot e^{(1-\alpha)\dot{n}/(\alpha\delta n_p)}$. Then:

$$\ln \tilde{\xi}(\dot{n}) = -\frac{1-\alpha}{2\alpha(\delta n_p)^2}\,\dot{n}^2 + O(\dot{n}^3).$$

This is $-\frac{\mu}{2}\dot{n}^2$ with

$$\mu = \frac{1-\alpha}{\alpha(\delta n_p)^2}.$$

Verification. For $\alpha = 0.5$ (symmetric matching), $\mu = 1/(\delta n_p)^2$. For $\alpha = 0.7$ (searcher-weighted), $\mu = 3/(7(\delta n_p)^2) \approx 0.43/(\delta n_p)^2$, approximately 57% smaller—implying faster cycles in industries with higher searcher matching power.

H Lindstedt-Poincaré Derivation of Frequency Correction

This appendix derives the $O(A^2)$ frequency correction for non-Gaussian profits stated in Proposition 4 of the main text.

Setup. Consider the oscillator $\ddot{x} + U'(x) = 0$, where $x = n - n_p$ and $U(x) = U_0 + \frac{1}{2}\omega_0^2 x^2 + \frac{1}{6}\beta x^3 + \frac{1}{24}\gamma x^4 + \cdots$, with $\omega_0 = \sqrt{U''(0)}$, $\beta = U'''(0)$, $\gamma = U^{(4)}(0)$.

Lindstedt-Poincaré expansion. Seek a periodic solution:

$$x(t) = A\cos(\omega t) + A^2 x_2(t) + A^3 x_3(t) + \cdots,$$

$$\omega = \omega_0 + A^2\omega_2 + O(A^4).$$

(Odd powers of A vanish by symmetry when $\beta = 0$.)

Order A^2. Substituting into the equation of motion and collecting $O(A^2)$ terms:

$$\ddot{x}_2 + \omega_0^2 x_2 = -\frac{\beta}{2}\cos^2(\omega_0 t) = -\frac{\beta}{4}[1 + \cos(2\omega_0 t)].$$

The particular solution is:

$$x_2(t) = -\frac{\beta}{4\omega_0^2} + \frac{\beta}{12\omega_0^2}\cos(2\omega_0 t).$$

Order A^3 (secular condition). At $O(A^3)$, the equation becomes:

$$\ddot{x}_3 + \omega_0^2 x_3 = \left[2\omega_0\omega_2 - \frac{\gamma}{8} + \frac{5\beta^2}{12\omega_0^2}\right]\cos(\omega_0 t) + \cdots$$

Eliminating the secular term (resonant forcing at frequency ω_0):

$$\omega_2 = \frac{1}{2\omega_0}\left[\frac{\gamma}{8} - \frac{5\beta^2}{12\omega_0^2}\right].$$

Result. The corrected frequency is:

$$\omega(A) = \omega_0\left[1 + \left(\frac{\gamma}{16\omega_0^2} - \frac{5\beta^2}{12\omega_0^4}\right)A^2 + O(A^4)\right].$$

For the Gaussian profit: $U(x) = \frac{\varphi}{2\mu}x^2$ exactly (no higher terms), so $\beta = \gamma = 0$ and $\omega(A) = \omega_0$ for all A (exact isochrony).

For the quartic perturbation $\pi(n) = \pi_0 e^{-\frac{\varphi}{2}x^2 - a_4 x^4}$: $U(x) = \frac{\varphi}{2\mu}x^2 + \frac{a_4}{\mu}x^4$, so $\beta = 0$ and $\gamma = 24a_4/\mu$. Then $\omega_2 = \frac{3a_4}{\mu\omega_0} = 3a_4/\varphi$ (using $\omega_0^2 = \varphi/\mu$), giving:

$$\omega(A) = \omega_0\left(1 + \frac{3a_4}{\varphi}A^2\right),$$

confirming the Duffing-oscillator formula used in the numerical experiments.

Cycle asymmetry (Proposition 5). When $\beta = U'''(0) \neq 0$, the $O(A^2)$ correction $x_2(t)$ includes a constant shift $-\beta/(4\omega_0^2)$ and a second-harmonic term. The constant shift displaces the time-averaged position away from n_p, making the orbit spend

unequal time above and below n_p. The leading-order asymmetry parameter is:

$$\mathcal{A} \equiv \frac{t_{\text{above}} - t_{\text{below}}}{t_{\text{below}}} = \frac{\beta}{4\omega_0^2} A + O(A^2).$$

Expressing β in terms of the profit function: $U'''(0) = -[\pi'''(n_p)\pi(n_p) - 3\pi''(n_p)\pi'(n_p)]/(\mu\pi(n_p)^2)$. Since $\pi'(n_p) = 0$, this simplifies to $\beta = -\pi'''(n_p)/(\mu\pi(n_p))$. Using $\omega_0^2 = |\pi''(n_p)|/(\mu\pi(n_p))$:

$$\mathcal{A} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|} A + O(A^2).$$

When $\pi'''(n_p) < 0$ (profit falls faster on the congestion side), $\mathcal{A} < 0$: the economy spends more time below n_p, producing slow expansions and sharp contractions. ■

I Stationary Distribution of the Damped Stochastic Oscillator

This appendix derives the stationary amplitude distribution stated in Proposition 7 of the main text.

Setup. The damped stochastic oscillator is:

$$\ddot{n} + \eta\,\dot{n} + \omega_0^2(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t,$$

where $\omega_0 = \sqrt{\varphi/\mu}$ and $\eta = \kappa/\mu$.

Energy equation. Define the energy $\mathcal{E} = \frac{1}{2}\dot{n}^2 + \frac{1}{2}\omega_0^2(n - n_p)^2$. Then:

$$\mathrm{d}\mathcal{E} = -\eta\,\dot{n}^2\,\mathrm{d}t + \frac{\sigma_\varepsilon}{\mu}\,\dot{n}\,\mathrm{d}W_t + \frac{\sigma_\varepsilon^2}{2\mu^2}\,\mathrm{d}t.$$

Stationarity condition. In stationarity, $\mathbb{E}[\mathrm{d}\mathcal{E}] = 0$:

$$\eta\,\mathbb{E}[\dot{n}^2] = \frac{\sigma_\varepsilon^2}{2\mu^2}.$$

Equipartition. For the linear damped oscillator with white noise, the stationary distribution of $(n - n_p, \dot{n})$ is bivariate Gaussian. Equipartition gives:

$$\mathbb{E}[\dot{n}^2] = \omega_0^2\, \mathbb{E}[(n - n_p)^2].$$

The Hilbert-envelope amplitude satisfies $\mathbb{E}[A^2] = 2\mathbb{E}[(n - n_p)^2]$. Therefore:

$$\eta \cdot \frac{1}{2}\omega_0^2\, \mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\mu^2}.$$

Solving:

$$\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{\mu^2\, \eta\, \omega_0^2} = \frac{\sigma_\varepsilon^2}{\mu\, \eta\, \varphi}.$$

Rayleigh distribution. The pair $(n - n_p, \dot{n}/\omega_0)$ is bivariate Gaussian with equal variances $\sigma_x^2 = \sigma_\varepsilon^2/(2\mu\eta\varphi)$. The envelope $A = \sqrt{(n - n_p)^2 + (\dot{n}/\omega_0)^2}$ therefore follows the Rayleigh distribution:

$$f(A) = \frac{A}{\sigma_x^2} \exp\left(-\frac{A^2}{2\sigma_x^2}\right), \qquad A \geq 0.$$

The mode is at σ_x and the mean is $\sigma_x\sqrt{\pi/2}$. Both scale linearly with σ_ε, confirming $A_{\text{rms}} \propto \sigma_\varepsilon$. ■